

Forecasting crude oil option prices with dynamic factors using integrated machine learning models

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ABSTRACT

In this article, we develop a dynamic factor model utilizing the incremental principal component analysis (IPCA) method to identify the key factors influencing crude oil option prices. We employ eight popular machine learning algorithms, i.e., LASSO, Support Vector Regression (SVR), Ridge Regression, Back Propagation (BP), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Multi-Layer Perceptron (MLP), to forecast crude oil prices. Their performance is rigorously evaluated using four distinct error measurements alongside the model confidence set (MCS) and direction of change (DoC) tests. We further introduce an integrated machine learning model that combines the strengths of the top-performing algorithms through weighted averaging, enhancing both predictive accuracy and robustness. In addition, we also explore the interpretability of the factors influencing the precision of predictivity using the SHAP value, so that the contribution of these factors towards the prediction is clear. Our findings offer substantial potential for mitigating energy risks and improving market efficiency through refined price forecasting techniques.

1. Introduction

There is a long history in the development of commodity derivatives globally, in which crude oil derivatives are one of the most important types. Crude oil is an important part of the energy structure of countries around the world, and its dramatic price fluctuations inevitably have a significant impact on the financial market, national economy and even national security [1]. The crude oil options market also plays a role in the energy sector by providing investors with an opportunity to trade crude oil in the future, and it is also an important tool for hedging risks. Accurate forecasting of crude oil option values is crucial for mitigating energy-related risks, a matter of significance to the government, investors, and regulators [2]. However, the crude oil market is influenced by several unpredictable factors, which are impacted by a range of elements such as the stock market, economic operations, political contexts, and other external influences [3], making it quite challenging to accurately predict the fluctuation of crude oil option prices.

Although crude oil futures and options of China were listed on the Shanghai International Energy Exchange quite recently, on

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March 26, 2018 and June 21, 2021, respectively, the trading volume of INE crude oil futures and options has grown rapidly. As China's first and currently Asia's only commodity futures and options market open to international investors [4], it is now the third largest crude oil futures and options market of the world. Although Asia holds the position of the largest consumer and importer of crude oil on a global scale, it has not developed an indigenous crude oil pricing framework, which can be attributed to the notorious "Asian Premium" [5]. Thus, the research on the related price discovery capability will help improve the pricing system of crude oil in China and even Asia.

Over the past decades, scholars have been working on developing various option pricing models, which can be broadly categorized into two forms: conventional parametric models and nonparametric models that are grounded in machine learning techniques [6]. The first category includes the Black-Scholes model [7] and the Heston model [8] as classical ones, and various more sophisticated ones, such as using wavelet technology [9], using indefinite differential equation [10], adopting a geometric Brownian motion and regime switching [11–13], and considering incomplete market conditions [14,15]. However, In the context of crude oil option price forecasting, it is important to note that Chinese crude oil options, as one type of American options, pose unique challenges that make traditional parametric models less effective. Unlike European options, which can only be exercised at maturity, American options can be exercised at any time before expiration. This flexibility significantly increases the complexity of pricing, as it requires the model to account for the potential early exercise of the option [16].

Given these complexities and the limitations of traditional models, there has been a growing interest in exploring the second category. Advancing in machine learning has led to a surge in research examining the potential of innovative artificial intelligence frameworks and data analysis techniques to bolster the precision and expedite the processes of valuation models. For example, Hutchinson et al. [17] first applied the neural network approach in machine learning to option price prediction, and they used the S&P 500 index futures and options as an empirical object. By comparing four model estimation methods, they derived the neural network model after repeated training, and proved that it outperformed the Black-Scholes model. Polanco-Martínez and Abadie [18] applied a stochastic model combined with wavelet analysis to explore the relationship between WTI crude oil spot prices and long-term futures prices, uncovering bidirectional causality across various time scales. Miao et al. [19] employed the LASSO regression technique to identify key factors influencing crude oil price forecasting, demonstrating a significant improvement in forecast accuracy over traditional benchmarks. In a subsequent study, Chai et al. [20] introduced a hybrid-refined method for forecasting WTI crude oil prices, integrating the PPM-KM model for change point detection, the TVTP-MRS model for regime switching identification, BMA for core factor selection, and the TVP-STSM for price estimation. This approach achieved superior forecasting performance compared to conventional models. Furthermore, Bredin et al. [21] showed that Nelson-Siegel factors derived from the WTI crude oil futures term structure effectively predict holding period returns, even when accounting for macroeconomic and oil-specific predictors. Their models, based on these factors, reduced forecast errors and enhanced trading strategy performance. More recently, Sen and Choudhury [22] employed advanced deep learning techniques, specifically LSTM and GRU models optimized using Particle Swarm Optimization (PSO), to forecast WTI crude oil prices over the past four decades. Their findings revealed that the PSO-GRU model outperformed the PSO-LSTM model.

It should be noted that these single machine learning models also have some limitations, such as parameter sensitivity and potential overfitting [23]. Moreover, machine learning models suffer from poor interpretability, as it is difficult to explain the decision logic behind the models, which limits their acceptability in practical applications. Our study aims to address these limitations by developing a dynamic factor model and integrating multiple machine learning algorithms to enhance predictive accuracy and robustness, while also providing interpretability of the factors influencing crude oil option prices.

The primary contributions of this paper can be summarized in three key areas. First, we develop a dynamic factor model that utilizes the incremental principal component analysis (IPCA) method. This approach allows us to identify the most significant factors and capture their potential changes in response to shifts in economic conditions, thereby improving the accuracy of our forecasts. Second, we introduce a novel integrated machine learning model that leverages the individual strengths of multiple models, aiming to strike a balance between stability and precision. This is accomplished through a carefully designed weighting scheme that ensures the model's reliability across different tests, significantly enhancing its collective predictive performance and robustness. Third, we use SHAP values to provide interpretability of the factors involved in the predictions, offering a deeper understanding of the key drivers influencing crude oil option prices.

The subsequent parts of this paper are organized as follows: Section 2 introduces the methodology used in this paper. Section 3 introduces and describes the data selection and indicator selection. Section 4 analyzes the results, and Section 5 draws relevant conclusions and policy implication.

2. Data

2.1. Variable selection

Based on the Wind database, this paper selects all the relevant data call options data since the listing of INE crude oil options, i.e., from June 21, 2021 to March 21, 2024. A total of 14858 pieces are collected, in which the training and test sets are divided by 10:1.

2.2. Dependent variable

Various options with diverse expiration dates and strike prices are traded simultaneously in the market. Typically, the most actively traded options are at-the-money ones. In particular, if K_t represents the strike price of the underlying futures, S_t represents the price of

the futures contract corresponding to the call option, and t denotes the date, then at-the-money options are referred to as those whose moneyness $|\frac{S_t - K_t}{K_t}|$ is no greater than 2.5%. Following Ahn et al. [24], we also focus on the option contract with the largest trading volume in a single day to represent the option price on that day, i.e.,

$$\begin{aligned} P_t &= \{P_t | \max(v_{t,i})\} \\ s.t. \quad & \left| \frac{S_t - K_t}{K_t} \right| \leq 2.5\% \end{aligned} \quad (1)$$

where P_t represents the selected option price and $v_{t,i}$ is the trading volume of the i th option on date t .

2.3. Independent variable

2.3.1. IPCA method

We now first identify a series of initial market characterization factors that are closely related to the options market. Considering the limitations of traditional time series analysis techniques, especially rolling window-based machine learning strategies, in capturing the dynamics of the options market in real time, we utilize the methodology developed by Büchner & Kelly [25] to downscale these characteristic variables with the IPCA method. Compared to the traditional principal component analysis, the IPCA is able to update factors by dividing the dataset into multiple small batches and processing these batches one by one, which enables the estimation of both fixed and varying factor loadings. This approach not only simplifies the process of calculating option risk characteristics, but also facilitates the construction of mathematical models between factor loadings and option characteristics. In view of this, IPCA shows its unique applicability in the construction of option pricing factor models [26].

The specific steps are as follows. We need to first calculate the covariance matrix S as

$$S = \frac{1}{N}(F - \hat{F})^T(F - \hat{F}) \quad (2)$$

where F is the matrix of initial market characteristics with dimensions $N \times M$ (with N samples and M market characteristics), and vector $\hat{F}$ is the mean vector of initial market characteristics with dimensions $1 \times M$. When calculating $F - \hat{F}$, the vector $\hat{F}$ is actually broadcast to an $N \times M$ matrix, with each row being $\hat{F}$ so that the subtraction can be conducted. As a result, $F - \hat{F}$ yields a centralized matrix with dimensions $N \times M$.

Then, an eigenvalue decomposition is performed on the covariance matrix S to obtain the eigenvalues λ_i and the corresponding eigenvectors v_i . In each iteration, the principal component Q is updated by incremental learning, and the update formula is as follows

$$Q_{t+1} = Q_t + \alpha \cdot (F_{batch} - \hat{F}_{batch})^T \quad (3)$$

where F_{batch} is the current batch, $\hat{F}_{batch}$ is the mean vector of the data in the current batch, and α is the learning rate.

By adopting this particular technique, we further construct a latent factor model on the prediction of option prices, which expresses the factor loadings as a function of observable option characteristics, thus providing a new perspective on option pricing.

2.3.2. Factor construction

As mentioned before, we choose to use the IPCA method for dimensionality reduction, based on which the selected initial market characteristic factors are used to construct 5 price prediction principal factors.

EPU (economic policy uncertainty) Factor: Crude oil options, as one type of financial derivatives, are subject to the combined influence brought by a variety of factors in terms of price volatility. Nowadays, the process of global integration is deepening and the links between different economies in the world are getting closer and closer, leading to a heightened sensitivity of the international crude oil market to the uncertainties in economic policies, both domestically and internationally. Both the research conducted by Dai et al. [27], as well as the study led by Wei et al. [28], have provided an in-depth examination of the influence that economic policy uncertainty exerts on the fluctuations within the crude oil market from the perspective of global integration. They come to the unanimous conclusion that the Economic Policy Uncertainty (EPU) index is a pivotal variable to be considered when analyzing the fluctuations of the crude oil market. Wei et al. [28] further pointed out that the EPU not only impacts the fundamentals of oil supply and demand, but also triggers speculative activities by changing the sentiment of market participants, thereby inducing volatility in the pricing of crude oil within the market.

The United States is recognized as a leading economic powerhouse globally, and changes in U.S. economic policies usually cause volatility and uncertainty in the global economy. Especially within the global crude oil market, the uncertainty of the U.S. economic policy will change investors' expectations of the global economy and crude oil demand, thereby exerting a subsequent influence on the pricing of crude oil options in China. In this paper, we consider U.S. EPU as a factor, measured using the EPU index, which was first pioneered by Baker et al [29].

MON (money market) Factor: With the increasing financialization of crude oil products and the advent of the era of big data, the phenomenon of information spillover from the financial sector to the crude oil market has grown increasingly pronounced [30]. The money market has become one of the most important channels for transmitting basic information about crude oil futures, and its interest rate changes reflect not only the adjustment of monetary policy, but also the official assessment of future economic conditions

[31]. The signals released by these interest rate movements directly affect investors' expectations of the RMB and the demand for crude oil, which in turn influences crude oil prices and consequently has an impact on the market for crude oil options to a certain extent. In this paper, we consider four currency market influences, including exchange rates, interest rates, Chinese Treasury rates, and U.S. Treasury rates.

Specifically, the exchange rate is ascertained based on the valuation of the RMB in comparison with the US Dollar. Meanwhile, the interest rate is formulated from the overnight lending rates within the Shanghai Interbank Offered Rate (SHIBOR) system and the Treasury rate is expressed as the one-year bond issuance rate in China and the United States respectively. Specifically, the exchange rate is ascertained based on the valuation of the RMB in comparison with the US Dollar. Meanwhile, the interest rate is formulated from the overnight lending rates within the Shanghai Interbank Offered Rate (SHIBOR) system.

ICOO (international crude oil options) Factor: In addition to the influence of international economic policies and financial markets, the interrelation between China's crude oil market and the international market is also an important factor. In their comprehensive study, Yang & Zhou [4] conducted an in-depth examination of the linkages between the Chinese crude oil market, specifically the International Exchange (INE), and its counterparts on the global stage, namely Brent and WTI crude oil futures markets. Their analytical findings revealed the existence of enduring price correlations and pronounced interplays in terms of returns and volatility dynamics between the INE and these two pivotal international benchmarks. Furthermore, the analysis also reveals a robust yet asymmetric relationship between the Chinese crude oil futures market and the predominant international crude oil futures markets. This characteristic of the correlation suggests that the Chinese market exhibits a heightened susceptibility to adverse price fluctuations emanating from its global counterparts, particularly during periods of market volatility. However, at present, the advancement of the INE has not supplanted the primary status held by WTI and Brent within the crude oil financial and pricing mechanisms [31]. The Volatility Index (VIX), often referred to as the "fear gauge," is an indicator of market turbulence constructed by the Chicago Board Options Exchange (CBOE). It is a commonly used measure in studies related to volatility forecasting in options [32]. Therefore, in this paper, five indicators, namely, VIX index, Brent crude oil price, WTI crude oil price, Brent crude oil future price, and WTI crude oil future price, are selected as international crude oil option market factors.

DCO (domestic commodity options market) Factor: Price fluctuations and trading activities in the domestic options market may also convey information that affects the expectations and behaviors of the China's crude oil options market. Supply and demand constitute the direct determinants of commodity price volatility, and thus shocks to crude oil supply and demand fundamentals will directly influence the volatility of crude oil futures prices [33]. In addition, some studies have confirmed the robust interconnection between the pricing of crude oil in the spot market and that of its futures contracts. Silvapulle and Moosa [34] suggested that futures prices typically align with spot prices upon contract delivery, and experimentally established the influence of spot crude oil prices on futures pricing. China Futures Commodity Index provides a comprehensive view covering a wide range of commodity categories from agricultural products to energy and metals. China Crude Oil Index, on the other hand, focuses on the energy market, in particular the fluctuation of crude oil prices, which is crucial for evaluating energy-related options. In addition, the CRB (Commodity Research Bureau) Spot Index, particularly its industrial component, reflects the supply and demand of industrial commodities, which is instructive for the domestic commodity options market. Therefore, these three indicators are selected.

CSM (Comprehensive stock market) Factor: Furthermore, an interconnection exists between equity market indices and the pricing of crude oil, characterized by a positive relationship between the directional movements of the stock and crude oil markets [35]. Moreover, speculative trading within the equity market is often found to invoke a temporary influence on the pricing of crude oil [36]. Crude oil and stock markets are also highly correlated in terms of risk levels, with significant risk spillover effects are observed between the crude oil and equity markets [37]. We considered three predictors that are representative of the Chinese and U.S. stock markets: the Shanghai Stock Exchange Composite Index, the S&P 500 Index, and the NASDAQ Composite Index. The volatility of the SSE Composite Index, as a representative of the Chinese equity market, directly affects the domestic stock options market. The S&P 500 and Nasdaq Composite, on the other hand, represent the performance of the U.S. stock market, especially technology stocks, and their

Table 1
Factor construction system.

Price prediction principal factors	Initial market characteristic	variable name
EPU Factor	Policy uncertainty index	EPU
MON Factor	Exchange rate	ER
	Interest rate	SHIBOR
	Chinese Treasury rate	CTR
	U.S. Treasury rate	USTR
	VIX index	VIX
ICOO Factor	Brent crude oil price	SPBRENT
	WTI crude oil price	SPWTI
	Brent crude oil future price	FPBRENT
	WTI crude oil future price	FPWTI
	China Futures Commodity Index	FC
DCO Factor	China Crude Oil Index	CO
	CRB Spot Index	CRB
	Shanghai Composite Index	SCI
CSM Factor	S&P 500 Index	S&P 500
	NASDAQ Composite Index	NASDAQ

movements have a non-negligible impact on the global stock market, including the Chinese options market. These indices not only reflect the macroeconomic conditions of their respective markets, but are also important tools for investors to assess market sentiment and formulate investment strategies.

As shown in Table 1, these factors generated through the IPCA dimension reduction method are intertwined and influence each other, and together shape the dynamics and development of the crude oil options market. Therefore, accurate prediction and analysis of crude oil option prices require comprehensive consideration of the influence of various factors and reasonable judgment and decision making in combination with specific circumstances.

2.2. Descriptive statistics

Fig. 1 displays that from June 2021 to March 2024, INE crude oil option prices experienced significant fluctuations. In June 2021, the price started from 20.75 and after an initial wobble, fell to a low of 17.45 in early July. In August, price volatility increased, reaching a high of 19.35 on August 5 before falling sharply. In October, prices jumped from 26.1 to 29.5 at the end of the month, showing a positive change in market sentiment. On November 29, the price rose sharply to 30.8 in an unusual manner, possibly due to market events. On March 2, 2022, the price reached the highest point of 71.85 in the observation period, after which it declined sharply. On January 4, 2023, the price rose again to a local high of 29.4 before starting to decline. On February 5, 2024, the price rose from a low value of 6.05 to 21.4, showing signs of recovery in the market. Overall, crude oil option prices were highly volatile during this period, showing the uncertainty and volatility of the market.

As shown in Table 2, the EPU index shows the high sensitivity of the crude oil market to changes in economic policy, with the mean reaching 129.00 and the standard deviation of 70.53 revealing its volatility. The maximum value of the EPU index at 1026.38 highlights the extreme uncertainty of the market in some periods. On the other hand, the mean value of ER is close to 6.81 with a small standard deviation, indicating a relatively stable exchange rate with low volatility. In terms of monetary policy, the average values of SHIBOR and CTR are 1.64 and 2.11, respectively, indicating a relatively stable interest rate environment in China. However, the higher standard deviation of the USTR indicates greater volatility in the US Treasury market, which may be related to global economic and political events. In terms of market volatility, the average value of VIX is as high as 62.71 with a standard deviation of 34.0356, which indicates that the market experienced high volatility during the study period and investors may face higher uncertainty and risk. In terms of crude oil prices, SPBRENT and SPWTI averaged to 86.68 and 82.69, respectively, while their futures prices FPBRENT and FPWTI showed a similar trend. These data indicate that the crude oil market is relatively stable during the study period, but there is still some volatility. In addition, commodity and stock market indices in China and globally, such as FC, CO, CRB, SCI, S&P 50 and NASDAQ, all show different volatilities and averages. In particular, the NASDAQ index has a mean value as high as 13327.00 and a standard deviation of 1638.60, indicating that the technology stock market may have experienced large fluctuations during the study period.

3. Methodology

3.1. Machine learning methods

In this study, we carefully select eight machine learning algorithms to forecast crude oil option prices, each chosen for its unique strengths and suitability to our data and prediction tasks. LASSO is selected for its feature selection capability via L1 regularization,

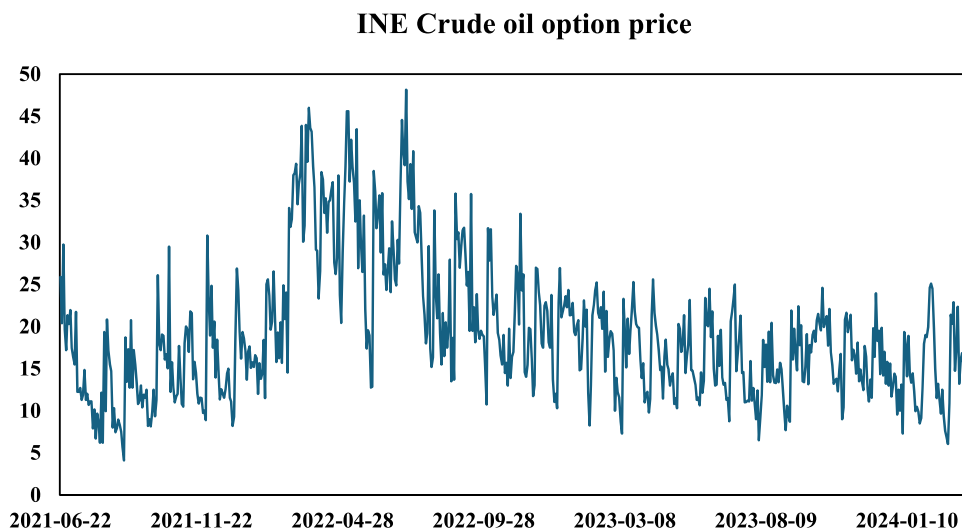


Fig. 1. Trend of crude oil option prices.

Table 2
Descriptive statistics of independent variables.

Variables	Count	Mean	Std	Min	25%	50%	75%	Max
EPU	645	129.00	70.53	14.69	88.12	116.73	155.79	1026.38
ER	645	6.81	0.31	6.30	6.47	6.88	7.11	7.26
SHIBOR	645	1.64	0.37	0.44	1.34	1.70	1.91	2.46
CTR	645	2.11	0.19	1.70	1.95	2.13	2.26	2.51
USTR	645	3.27	2.04	0.06	1.11	4.51	5.00	5.49
VIX	645	62.71	34.04	13.00	25.55	68.55	88.45	142.80
SPBRENT	645	86.68	12.20	65.18	77.72	83.72	93.17	127.98
SPWTI	645	82.69	12.46	62.14	73.32	79.38	88.37	123.70
FPBRENT	645	86.98	12.46	65.18	77.99	83.72	93.26	127.98
FPWTI	645	82.72	12.44	62.32	73.47	79.46	88.32	123.70
FC	645	2280.74	211.87	1853.49	2145	2303.08	2467.22	2625.6
CO	645	904.76	152.84	564.43	809.18	933.10	1014.76	1188.19
CRB	645	591.86	44.20	535.16	554.29	572.65	635.51	686.18
SCI	645	3251.12	212.81	2702.19	3085.98	3237.54	3409.21	3715.37
S&P500	645	4313.41	337.32	3577.03	4045.64	4352.63	4522.79	5241.53
NASDAQ	645	13327.00	1638.60	10213.29	11816.2	13542.12	14714.66	16401.84

which effectively shrinks less important features to zero, thereby enhancing model interpretability. SVR is chosen for its robustness in handling high-dimensional and non-linear data, leveraging support vectors and kernel functions to minimize overfitting. Ridge Regression is included for its ability to address multicollinearity and overfitting through L2 regularization, making it well-suited for datasets with highly correlated predictors. BP, a fundamental feedforward neural network, is selected for its capacity to learn complex patterns through its layered structure and non-linear transformations. RNN, LSTM, and GRU are chosen for their effectiveness in processing sequential data and capturing temporal dependencies, with LSTM and GRU specifically designed to address long-term dependencies and improve on the limitations of traditional RNNs. Finally, MLP is included for its versatility in handling non-linear problems and its ability to approximate arbitrary continuous functions through its multiple layers of neurons. The details on the eight machine learning methods are provided in [Appendix 1](#) for the completeness of the paper.

3.2. The integration of machine learning models

Rapach et al. [38] discovered that the forecasting performance of the equity premium can be improved through combined forecasts. As a result, we adopt a similar weighted average approach and combine the forecast results of multiple machine learning models to form a combined forecast of option prices. Specifically, the combined forecast of $\hat{p}_{c,t+1}$ made at time t is obtained based on the weighted average of the forecasts $\hat{p}_{i,t+1}$ of N independent models, i.e.,

$$\hat{p}_{c,t+1} = \sum_{i=1}^N \omega_{it} \hat{p}_{i,t+1} \quad (4)$$

The corresponding weights are determined by the prediction performance of each model's historical out-of-sample performance:

$$\omega_{it} = \frac{\left[\sum_{s=m}^{t-1} \theta^{t-1-s} (p_{s+1} - \hat{p}_{i,s+1})^2 \right]^{-1}}{\sum_{j=1}^N \left[\sum_{s=m}^{t-1} \theta^{t-1-s} (p_{j,s+1} - \hat{p}_{j,s+1})^2 \right]^{-1}} \quad (5)$$

Among them, a larger weight is assigned to the prediction of a single predictive regression model with a lower MSPE value during the retention out-of-sample period, and the Discounted Mean Squared Prediction Error (DMSPE) method is adopted for the weight. When $\theta = 1$, there is no discounting. When $\theta < 1$, more weight will be placed on the forecast accuracy of each model in the near future.

Further, we selected the four machine learning models with the best performance (MSE evaluation value and out-of-sample fit performance) from the previous experiments and weighted them. This strategy aims to enhance the stability of forecasts and improve the accuracy when forecasting option price dynamics through integration. In this way, we are able to enhance the adaptability and robustness of the model to market changes while maintaining accuracy.

3.3. Evaluation Methods

3.3.1. Loss Functions

The evaluation of machine learning models is particularly critical in the field of option price prediction because of the high complexity in the dynamics of the option market. MAE is favored in the quantification of option pricing errors due to its computational simplicity and robustness to outliers. Nevertheless, MAE does not consider the direction of the error, which can be a limiting factor in applications. MSE and RMSE, on the other hand, provide absolute measures of error by squaring the penalty for larger errors and are statistically excellent in nature [39]. RMSE has the advantage of being on the same scale as the original data, which makes the interpretation of error more intuitive. However, both metrics are very sensitive to outliers and may require additional data processing

steps to mitigate the effects of extreme values. MAPE presents the relative error between the predicted value and the actual value in percentage form, and is particularly suitable for investors to understand how the model performs at different option price levels. MAPE can become unstable at smaller actual values, and therefore it needs to be used with caution when option prices are volatile [40]. The corresponding definition of these four indicators are given as

$$MAE = \frac{1}{N} \sum_{t=M+1}^{M+N} \sum_{*} |\hat{p}_t^{*} - p_t^{*}| \quad (6)$$

$$MSE = \frac{1}{N} \sum_{t=M+1}^{M+N} \sum_{*} |\hat{p}_t^{*} - p_t^{*}|^2 \quad (7)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=M+1}^{M+N} \sum_{*} |\hat{p}_t^{*} - p_t^{*}|^2} \quad (8)$$

$$MAPE = \frac{1}{N} \sum_{t=M+1}^{M+N} \sum_{*} \frac{|\hat{p}_t^{*} - p_t^{*}|}{p_t^{*}} \times 100\% \quad (9)$$

It should be emphasized that a combination of these metrics needs to be considered for assessing the performance of price forecasting. For example, if the model is intended for trading strategies, more attention may be paid to the RMSE so that large pricing errors can be captured. Moreover, if it is intended for risk management, the MAE may be preferred to assess the level of average error. In addition, MAPE may be more suitable for those scenarios where it is necessary to evaluate the forecasting performance of the model at different price intervals.

3.3.2. MCS tests

In the evaluation process of option pricing models, while traditional error metrics can reveal the predictive accuracy of the model, they are often insufficient to comprehensively evaluate the predictive efficacy of the model. To overcome this limitation, we introduce the MCS test, which is a more comprehensive method for comparing model performance.

The MCS test identifies and compares models with optimal predictive performance by constructing a confidence set containing all candidate models [41]. The method first creates a set M_0 containing n predictive models, and then progressively reduces the set of models through a series of statistical tests until an optimal set of models M^* is identified. The MCS test centers on two steps: equivalence test and model exclusion. The null hypothesis of the equivalence test is that all candidate models are equivalent in terms of the predictive performance, while the exclusion rule is used to eliminate poorly performing models when the null hypothesis is rejected. This process involves evaluating the statistical significance of the models' forecasting errors to ensure that the models in the final selected set of models M^* are significantly better at forecasting option prices.

We apply the mean loss variance to compare errors of different models and overall significance test. The formula is given by

$$\bar{d}_{ij,t} = \frac{1}{T} \sum_{t=1}^T l_{i,t} - l_{j,t} \quad (10)$$

where $l_{i,t}$ and $l_{j,t}$ are the losses of the i -th and j -th models at time t , respectively. They are used overall to estimate the average performance difference between model i and model j . We then treat the p-value of the above MCS test as an important measure for the predictive performance of the model; the closer the p-value of the MCS test is to 1, the stronger the predictive performance of the model, and thus the more reliable the model is in predicting the price of an option.

In the field of option pricing, the MCS test provides a mechanism for identifying models that maintain stable predictive performance under changing market conditions. Through the MCS test, option pricing models that are best suited to specific market conditions can be more accurately evaluated and selected, providing a more solid basis for decision-making in option trading and risk assessment.

3.3.3. DoC tests

The DoC metric is used to assess the accuracy of the model in predicting the direction of option price changes [42]. We assume that p_t is a dummy variable. Its value should be 1 if the prediction model accurately determines the direction of price change on day t , and 0 otherwise, i.e.,

$$p_t = \begin{cases} 1, & \text{if } x_t^* > x_{t-1}^* \text{ and } \hat{x}_t^* > x_{t-1}^* \text{ or } x_t^* < x_{t-1}^* \text{ and } \hat{x}_t^* < x_{t-1}^* \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

This means that $p_t = 1$ if both of the actual and predicted prices, x_t^* and $\hat{x}_t^*$, are higher (lower) than those of the previous day x_{t-1}^* , respectively, and $p_t = 0$ means that the prediction is in an opposite direction.

The DoC metric provides a quantitative evaluation tool for option pricing models to visualize how accurate the model is in predicting the direction of market price changes. A high DoC value indicates that the model not only has good predictive power, but is also

likely to be more reliable in actual trading, thus enhancing the effectiveness of trading strategies and risk management. By continually monitoring and analyzing DoC values, one can continuously optimize the forecasting models to adapt to the rapidly changing market environment.

3.3.4. DM tests

The Diebold-Mariano (DM) test is a widely used statistical method for comparing the predictive accuracy of different models [43]. It is particularly useful in the context of option price forecasting, where the goal is to determine whether the differences in prediction errors across models are statistically significant. The DM test focuses on the comparison of the forecast errors of two models, allowing for a more nuanced understanding of their relative performance.

In this study, we applied the DM test to compare the predictive accuracy of the individual machine learning models and the weighted model. The test statistic is based on the difference in the forecast errors of the two models being compared, and it follows a normal distribution under the null hypothesis of no difference in predictive accuracy. The DM test statistic is calculated as follows:

$$DM = \frac{\bar{d}}{\sqrt{\frac{Var(d_t)}{T}}} \quad (12)$$

where $\bar{d}$ is the mean of the differences in forecast errors between the two models, d_t is the difference in forecast errors at time t , $Var(d_t)$ is the variance of the differences in forecast errors, and T is the number of observations.

3.4. SHAP value

The main idea of the SHAP value was derived from the coalitional game theory and proposed by Shapley [44]. It was applied to the domain of feature importance by Lundberg & Lee [45]. Its biggest advantage is that it can reflect the influence of characteristics in each sample, as well as their positive and negative effects. The proposed method can extract and visually depict both the features that contributed to the outlier most and those that offset it [46].

The prediction of the machine learning model for each instance can be reproduced as the sum of these SHAP values plus fixed fundamental values, making the model output,

$$f(x) = \text{base value} + \text{sum}(\text{SHAP values}) \quad (13)$$

The Shapley value of each eigenvalue is the contribution of that eigenvalue to the prediction, obtained by summing up the weighted sum of all possible combinations of eigenvalues,

$$\phi_j(\text{val}) = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|!(p - |S| - 1)!}{p!} (\text{val}(S \cup \{x_j\}) - \text{val}(S)) \quad (14)$$

where $\phi_j(\text{val})$ (SHAP value) is the attribution value of each feature, S is the subset (coalition) of features used in the model, x is the vector of feature values of the instances to be explained, p is the number of features. $\frac{|S|!(p - |S| - 1)!}{p!}$ is the weight of subset S , and this weight depends on the order. $\text{val}_x(S)$ is the prediction of subset S and is marginalized on features not included in the set S :

$$\text{val}_x(S) = \int \hat{f}(x_1, \dots, x_p) dP_{x \notin S} - E_X(\hat{f}(X)) \quad (15)$$

Feature importance is measured by averaging the absolute Shapley value of each feature, with higher average absolute SHAP values indicating more influential features.

$$I_j = \sum_{i=1}^n |\phi_j^{(i)}| \quad (16)$$

4. Empirical analysis

4.1. Selection of important parameters

The sliding window technique is applied for predictive modeling in machine learning. We compare the performance of the models using four different sliding window lengths: 5, 10, 15, and 20 days. The results in Table 3 shows that the model with a 10-day sliding window yields the best performance, and thus we will use the 10-day sliding window in the following comparison.

Besides, in the integrated machine learning model, regarding the determination of θ , we employed a grid search method to systematically explore the values of theta from 0.1 to 1, incrementing by 0.1 for each step [38]. Fig. 2 indicates that the MAE remains relatively stable with respect to theta, while the MSE is minimized when θ is set to 0.9. Therefore, $\theta = 0.9$ will be used in the following experiments. Other optimal parameters associated with the used machine learning methods are provided in Appendix 2.

4.2. The model performance

To present the model's effectiveness in a clearer manner, we graphically depicted the assessment criteria, taking into account the complete spectrum of contributing factors, as shown in Fig. 3.

From the results, the LASSO model demonstrates a strong performance with a MAE of 2.67, an MSE of 11.01, an RMSE of 3.32, and a MAPE of 0.19, indicating its effectiveness in predicting crude oil option prices. The SVR model, while showing good prediction accuracy, has a slightly higher MAE of 3.00, an MSE of 13.86, an RMSE of 3.72, and a MAPE of 0.20, suggesting some limitations in handling outliers. The Ridge model performs similarly to SVR, with a MAE of 2.82, an MSE of 13.22, an RMSE of 3.64, and a MAPE of 0.18, showing its capability in addressing multicollinearity and overfitting. The BP model exhibits a moderate performance, with a MAE of 3.13, an MSE of 13.79, an RMSE of 3.71, and a MAPE of 0.24, which may be attributed to its slower learning rate or tendency to fall into local minima. The RNN model shows a slight improvement, with a MAE of 2.84, an MSE of 13.60, an RMSE of 3.69, and a MAPE of 0.19, demonstrating its potential for processing sequential data. The LSTM model, despite its theoretical advantage in learning long-term dependencies, performs slightly worse than RNN, with a MAE of 2.78, an MSE of 12.70, an RMSE of 3.56, and a MAPE of 0.18. In contrast, the GRU model outperforms LSTM, with a MAE of 2.79, an MSE of 12.83, an RMSE of 3.58, and a MAPE of 0.19, indicating its effectiveness in handling the dataset. The MLP model shows excellent performance, with a MAE of 3.08, an MSE of 15.80, an RMSE of 3.98, and a MAPE of 0.21, highlighting its superior performance and wide applicability.

Meanwhile, The Weighted model, which is composed of the four models with the lowest MSE (LASSO, GRU, Ridge, and LSTM), demonstrates superior predictive capabilities and robustness. Specifically, the LASSO model contributes with a MAE of 2.67, an MSE of 11.01, an RMSE of 3.32, and a MAPE of 0.19, highlighting its effectiveness in feature selection and handling multicollinearity. Our evaluation results show that the weighted model achieves significant improvements in several key performance metrics when compared to a single model, especially when using the MLP model as a reference.

Specifically, the weighted model achieves a reduction of approximately 3.37% in MAE compared to the LASSO model, which significantly improves the model's prediction accuracy. For options trading, this improvement is critical because a lower MAE value means that there is less discrepancy between predicted and actual prices, which can more accurately guide trading decisions. In terms of RMSE, the weighted model also demonstrates an improvement of approximately 10.16% compared to the Ridge model, which further confirms its strength in predictive stability and reliability. Although the weighted model performs comparably to the LSTM model in terms of MSE, this does not diminish the excellent performance it exhibits in other metrics. In addition, the weighted model achieves an improvement of approximately 5.56% in MAPE compared to the GRU model, which highlights its improvement in relative error, especially in terms of accuracy in predicting percentage changes in prices. Overall, the weighted model's superior performance in multiple metrics underscores its effectiveness in integrating the strengths of individual models to enhance forecasting accuracy and robustness.

As shown in Fig. 4, in terms of outliers, we can also find that in the box plot of error analysis (the vertical axis represents the difference between the predicted value of the model and the true value), the weighted model performs best, and the predicted value is closest to the true value with a small bias.

The introduction of the weighted model in Fig. 5 significantly improves the forecasting performance of the option pricing model, not only improving the accuracy of the forecasts, but also enhancing the model's adaptability to market volatility, ensuring that it maintains a high level of performance in different market environments.

Based on the data provided in Table 4, we observe that the MCS P-values of the SVR, and BP are all 0, which indicates that these models differ significantly from the others in terms of their predictive performance, and cannot be considered equivalent. Meanwhile, the MCS P-values of LASSO, Ridge, RNN, LSTM, GRU, MLP, and Weighted model are all 1, which may mean that these models do not show significant differences in prediction performance or that there is not enough data to reject their null hypothesis of equivalence.

Further, when we consider the DoC test results in Table 5, we find that these models are more accurate and statistically significant in predicting the direction of price changes. In particular, the weighted model has a DoC P-value of 0.9921, indicating a very high accuracy in predicting direction.

The Diebold-Mariano (DM) test is employed to statistically compare the predictive accuracy of the individual models with the weighted model. The test results are presented in Table 6. The results reveal that the weighted model significantly outperforms most of the individual models in terms of predictive accuracy. Specifically, the p-values for the comparisons between the weighted model and SVR, Ridge, BP, RNN, LSTM, GRU, and MLP are all below the conventional significance level of 0.05, indicating that the differences in predictive accuracy are statistically significant. The weighted model shows a substantial improvement over these models, as evidenced by the low p-values. For instance, the DM test statistic for SVR vs. Weighted is 3.75 with a p-value of 0.0003, suggesting that the weighted model has a significantly higher predictive accuracy than the SVR model. In contrast, the LASSO vs. Weighted comparison has a DM statistic of 0.479 and a p-value of 0.6326, indicating that there is no statistically significant difference in predictive accuracy

Table 3
The MSE values of each model under different sliding window lengths.

Length	Lasso	SVR	Ridge	BP	RNN	LSTM	GRU	MLP	Weighted
5	12.59	13.61	13.97	16.20	14.36	24.96	13.52	21.00	12.39
10	13.65	15.02	14.77	15.02	14.75	17.37	13.01	23.67	11.40
15	15.34	23.54	19.74	24.30	14.78	29.38	12.60	29.66	14.28
20	14.85	21.34	19.18	17.54	13.65	33.08	11.83	61.90	11.81

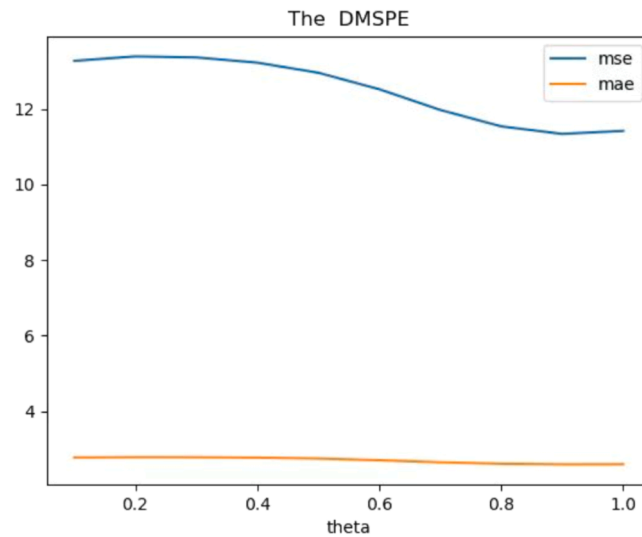


Fig. 2. The nonlinear relationship between θ and MSE as well as MAE.

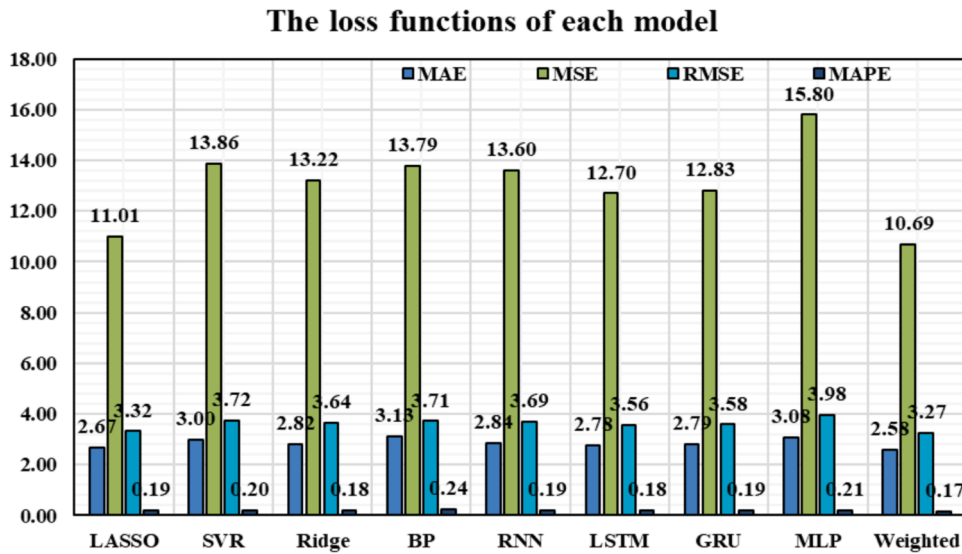


Fig. 3. The loss function of each model.

between the LASSO model and the weighted model. This suggests that while LASSO performs reasonably well, it does not offer a significant improvement over the weighted model.

We further investigate whether the inclusion of the lagged EPU factor would significantly alter the performance of the models in Table 7. The small variations in performance metrics across models indicate that the lagged EPU factor has a minimal impact on the predictive accuracy of crude oil option prices. This suggests that while EPU is an important factor, its lagged effect does not significantly affect the model's performance. As a result, the model's robustness is preserved, and its predictive accuracy remains consistent, even when accounting for the lagged EPU factor.

4.3. Factor interpretability

Figs. 6 and 7 show that the ICOO Factor has the highest SHAP value of +4.58 among the five factors, and the distribution shows obvious clustering with high values in most cases. This indicates that it has the most significant influence on predicting crude oil option prices and such influence is significantly positive, demonstrating that the international crude oil market is vital to INE crude oil options. One can also find that the CSM Factor plays a significant role in affecting the model prediction, although its effects are not as large as the ICOO Factor. In contrast, the SHAP values of the DCO Factor and MON Factor are similar at +1.09 and +1.04, respectively,

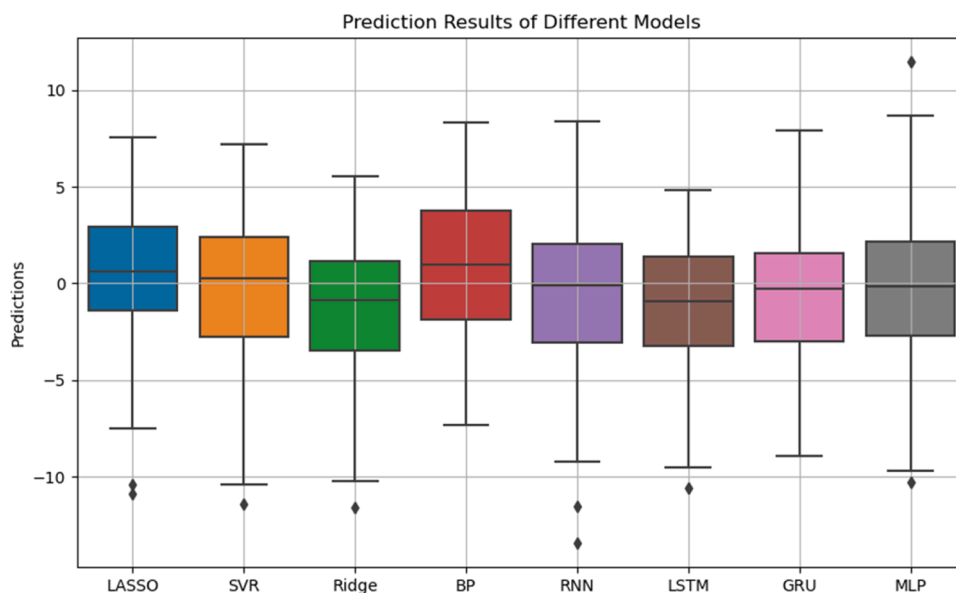


Fig. 4. Box plot of the bias distribution.

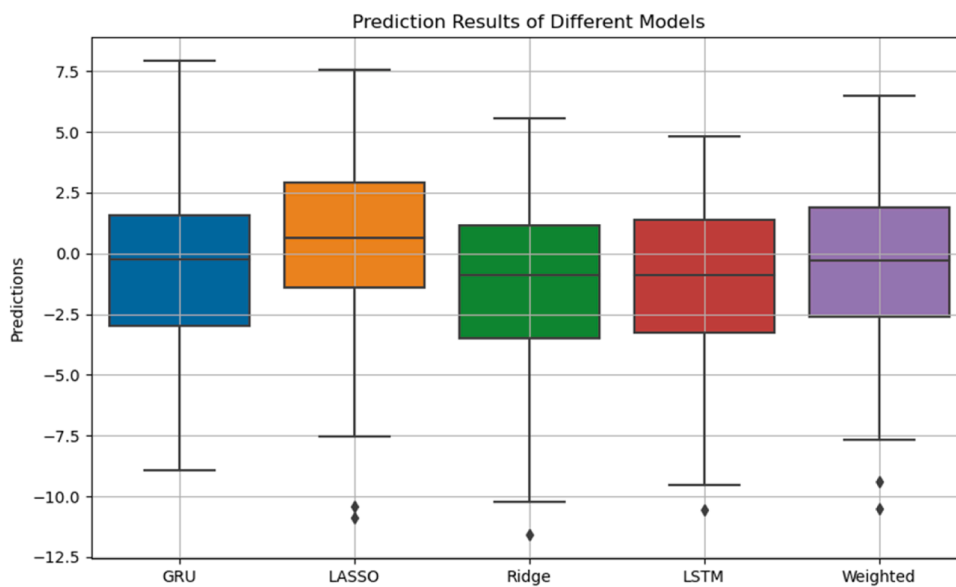


Fig. 5. Box plot of the bias distribution in the weighted model.

Table 4
MCS tests.

Models	MCS P-values
LASSO	1
SVR	0
RIDGE	1
BP	0
RNN	1
LSTM	1
GRU	1
MLP	1
Weighted	1

Table 5

DoC tests.

MODELS	DoC tests
Lasso	0.5238
SVR	0.4683
Ridge	0.4524
BP	0.5238
RNN	0.4286
LSTM	0.4921
GRU	0.5397
MLP	0.5476
Weighted	0.9921

Table 6

DM tests.

Models Compared	DM Statistic	p-value
LASSO vs. Weighted	0.479	0.6326
SVR vs. Weighted	3.75	0.0003
Ridge vs. Weighted	3.602	0.0005
BP vs. Weighted	2.282	0.0241
RNN vs. Weighted	2.855	0.005
LSTM vs. Weighted	2.736	0.0071
GRU vs. Weighted	2.76	0.0066
MLP vs. Weighted	3.531	0.0006

Table 7

Model Performance with Lagged EPU Factor.

	BP	MLP	RNN	LSTM	GRU	Lasso	Ridge	SVR	Weighted
MAE	4.68	3.82	3.45	4.83	3.03	2.84	2.99	3.05	2.63
MSE	34.18	26.40	18.96	32.01	14.33	13.65	14.94	15.49	11.63
RMSE	5.85	5.14	4.36	5.66	3.79	3.69	3.87	3.94	3.41
MAPE	0.29	0.25	0.23	0.34	0.22	0.18	0.18	0.19	0.17

and their distributions are more dispersed and concentrated on the lower numerical region. This may reveal that the overall influence of both factors on the model is weak. Although the EPU Factor has an impact on the market sentiment and presents a balanced distribution, its direct impact on crude oil option prices is relatively low, with the SHAP value of +0.98.

The results of feature hierarchical clustering also reveal possible links between different factors. One can identify some connection between the financial market related factors, such as the CSM Factor and DCO Factor, and economic policy uncertainty (EPU). This relationship helps us better understand market dynamics and how various factors together affect crude oil option prices.

In Fig. 8, both the ICOO Factor and CSM Factor show a significant positive contribution to the model output, with the ICOO Factor significantly pushing up the predicted value in most cases, which further reflects the significant impact of international market conditions on crude oil option prices. In contrast, although the impact of EPU Factor is complex, showing bilateral effects in both positive and negative directions, it tends to have a slightly positive impact on the whole, revealing that the economic policy uncertainty may increase the market demand for crude oil options as a risk hedging tool in some cases. Although the MON Factor and DCO Factor are also involved in the model decision, their impact is relatively small. This demonstrates that changes in the money market and domestic commodity market have less significant impact on crude oil option prices than international factors.

5. Conclusion and policy implication

In this study, we adopt the IPCA technique to effectively downscale the initial market characteristic variables during the model construction process. Based on the obtained dynamic factor model, eight machine learning models are adopted to predict crude oil option prices. We also establish an integrated machine learning model, which further improves the accuracy and robustness of the prediction, with the improvement being 7% on average. The integrated model has a p-value of 1.0 in the MCS test, showing that it is statistically significantly better than the other models. It also demonstrates a very high accuracy in the DoC test, with a p-value as high as 0.9921, which further validates its superiority in predicting the direction of price movements. The DM test results reveal that the weighted model significantly outperforms most of the individual models in terms of predictive accuracy.

The comparison between our study and Miao et al. [19] highlights the differences in the choice of machine learning methods and the specific focus on option prices rather than spot prices. While LASSO regression is effective in selecting key factors and improving forecasting accuracy for spot prices, our study demonstrates that an integrated machine learning approach can further enhance the

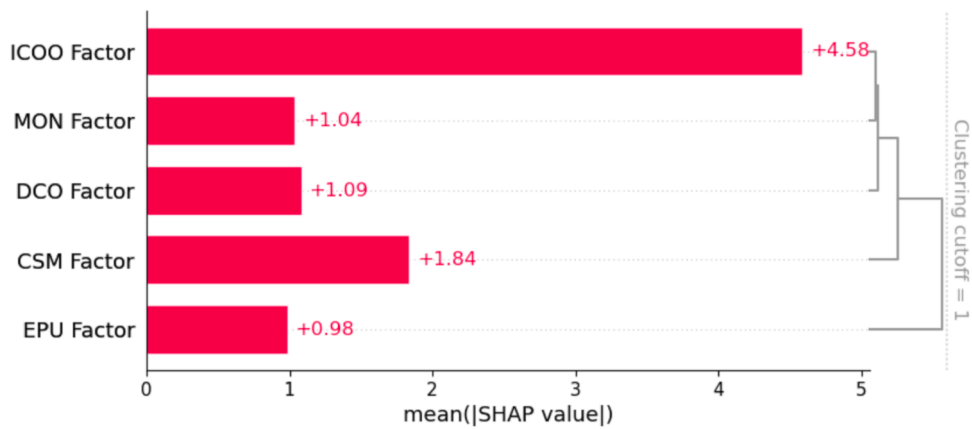


Fig. 6. Hierarchical clustering histogram of factor importance.

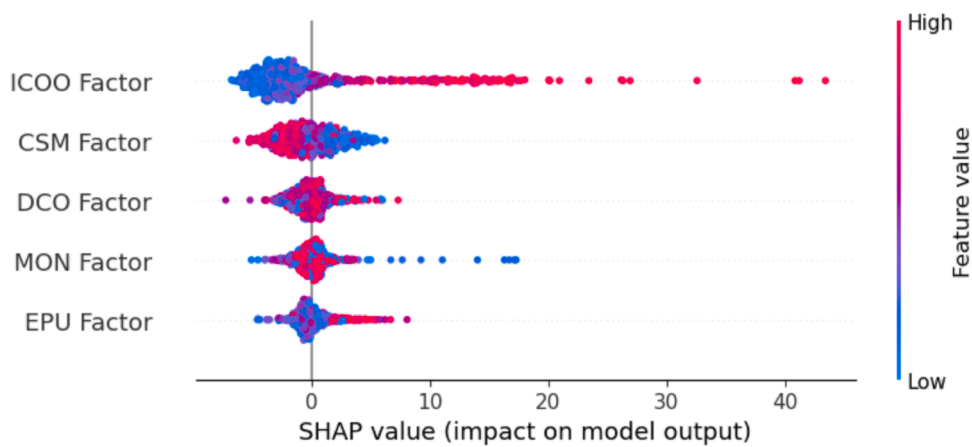


Fig. 7. SHAP hive diagram of factors.

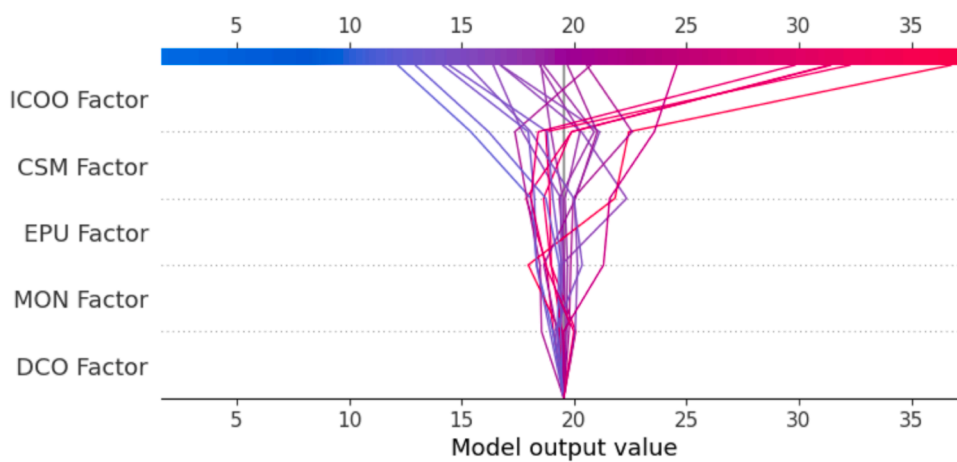


Fig. 8. SHAP decision tree diagram.

predictive performance for option prices. This is particularly evident in the context of the highly volatile and complex crude oil market, where the integration of multiple machine learning models provides a more comprehensive and robust solution.

Moreover, our study highlights the significance of international market conditions, as captured by the ICOO factor, in shaping

crude oil option prices. This finding aligns with Miao et al. [19], who concluded that international factors are crucial drivers of crude oil price fluctuations. However, our research goes further by incorporating a broader set of market factors, including economic policy uncertainty, financial markets, and commodity markets, into a dynamic factor model. Additionally, we find that the DCO and MON factors have a lesser impact, suggesting that the money market and domestic commodity markets exert minimal influence on crude oil option prices.

Our study emphasizes the potential of machine learning techniques in improving the efficiency of financial derivatives pricing. Building on these insights, we propose several policy recommendations. First, policymakers and regulators should prioritize the understanding and responding to international market dynamics by developing comprehensive risk management frameworks that mitigate the effects of global market fluctuations. Second, integrating the IPCA method and machine learning into regulatory practices can improve the monitoring and regulatory frameworks, allowing for a proactive approach to managing market conditions. Third, the financial sector should invest in fintech innovations that enhance predictive accuracy and facilitate better decision-making, thus stabilizing and optimizing market operations. Lastly, we advocate for increased educational efforts to enhance the transparency and comprehensibility of these complex models, fostering greater trust and participation among market players. Collectively, the adoption of advanced analytical tools coupled with a focus on educational initiatives offers substantial benefits for financial regulation and market stability. In summary, our study not only introduces innovative perspectives and methods for crude oil option pricing but also substantiates the application and advancement of fintech through empirical evidence. By leveraging the IPCA method and machine learning, we have enhanced the interpretability of significant market factors, thus providing deeper insights into the drivers of crude oil option prices. The analysis of factor interpretability, particularly the dominant influence of the ICOO factor, offers a deep understanding on how global market conditions shape pricing dynamics. Future research can aim to explore variations in model performance under diverse market conditions and investigate the integration of macroeconomic indicators with other financial market data. This would enhance the generalization capability and adaptability of forecasting models, further refining our understanding of market mechanisms and influences.

CRediT authorship contribution statement

Kexin Yu: Writing – original draft, Software, Investigation, Formal analysis. **Xin-Jiang He:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Xiaoyang Han:** Writing – review & editing, Software. **Xin Luo:** Writing – review & editing, Validation. **Sha Lin:** Writing – review & editing, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Credit author statement

Kexin Yu: Formal analysis, Investigation, Software, Writing- Original draft preparation. **Xin-Jiang He:** Conceptualization, Methodology, Supervision, Validation, Writing- Reviewing and Editing. **Xiaoyang Han:** Software, Writing- Reviewing and Editing. **Xin Luo:** Validation, Writing- Reviewing and Editing. **Sha Lin:** Methodology, Software, Writing- Reviewing and Editing.

Appendix 1: Eight machine learning methods

1 LASSO

LASSO (Least Absolute Shrinkage and Selection Operator) regression is a linear regression technique that incorporates L1 regularization to enhance model performance by shrinking the coefficients of less important features to zero. This characteristic makes LASSO particularly useful for feature selection, especially in scenarios where the dataset contains a large number of potentially irrelevant or redundant variables. In the context of crude oil option price prediction, LASSO can help identify the most influential factors by penalizing the absolute size of the regression coefficients, thus simplifying the model and improving its interpretability.

The objective function of LASSO regression is defined as:

$$\min \left[\frac{1}{2N} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right] \quad (17)$$

where y_i represents the dependent variable (crude oil option prices), x_{ij} are the five independent market factors, β_j are the coefficients, λ is the regularization parameter controlling the strength of the penalty, and N is the number of observations. By adjusting λ , LASSO can balance the trade-off between model complexity and accuracy, ensuring that only the most relevant factors are retained in the final model. This feature is particularly advantageous in financial markets, where the relationship between variables can be highly nonlinear and influenced by numerous external factors.

2 SVR

In SVM regression analysis, the construction of the model relies heavily on a subset of the training data, i.e., the support vectors. Denote a training set as $\{(x_1, y_1), \dots, (x_l, y_l)\} \subset X \times R$, where X represents a multidimensional space defined by the input parameters (e.g., $X = R^d$), such as the price of the asset and the price of execution. The goal of the SVM regression is to minimize the slope of the function $f(x)$ as much as possible [47] under the condition that the bias is no more than $1/2$. This particular condition means that the difference between the model's predicted value and the true value is acceptable within a certain range. Reducing the slope of the function is to make the model smoother and avoid overfitting, thus improving the generalization ability of the model, i.e.,

$$\min f(x) = \left[\epsilon \|\omega\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \right] \quad (18)$$

$$\text{s.t.} \begin{cases} y_i - \langle \omega, x_i \rangle - b \leq \epsilon + \xi_i \\ \langle \omega, x_i \rangle + b \leq \epsilon + \xi_i \\ \xi_i, \xi_i^* \geq 0 \end{cases} \quad (19)$$

where $\epsilon = \frac{1}{2}$ defines the tolerance bounds of the optimization problem, and the constant $C > 0$ balances the smoothness of the function f with the fit to the training data.

When dealing with nonlinear problems, the input space is mapped to a higher dimensional feature space by means of a kernel function $k(x_i, x)$, such as the radial basis function (RBF) kernel $k(x_i, x) = e^{-\gamma \|x_i - x\|^2}$. Further, the dyadic optimization problem for solving the SVR can be obtained with the Lagrange multiplier method and the construction of the dyadic problem. In the dyadic form, the Lagrange multipliers α_i and α_i^* are introduced and an expression for the regression function $f(x)$ can be obtained by solving these multipliers. The final regression function will depend only on those support vectors that correspond to non-zero Lagrange multipliers.

SVR has been used in a variety of regression analysis scenarios due to its ability to find optimal hyperplanes in high-dimensional spaces and its ability to handle nonlinear problems via kernel functions. In addition, the robustness of SVR gives it an advantage on datasets with noise and outliers.

3 Ridge Regression

Ridge regression, another regularization technique, employs L2 regularization to address multicollinearity and overfitting in linear regression models. Unlike LASSO, which can shrink coefficients to zero, Ridge regression shrinks the coefficients of correlated predictors towards each other, distributing the effect among them. This makes Ridge regression particularly suitable for situations where predictors are highly correlated, as is often the case in financial markets where factors such as interest rates, exchange rates, and commodity prices may exhibit strong interdependencies.

The objective function of Ridge regression is given by:

$$\min \left(\frac{1}{2N} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right) \quad (20)$$

where λ is the regularization parameter that controls the amount of shrinkage applied to the coefficients. By penalizing the square of the coefficients, Ridge regression ensures that the model remains stable and less sensitive to small changes in the input data. This stability is crucial in the context of crude oil option price prediction, where market conditions can be highly volatile and influenced by a wide range of economic and geopolitical factors. Ridge regression's ability to handle multicollinearity makes it a valuable tool for capturing the complex relationships between market variables, thereby improving the robustness of the predictive model.

4 BP

A BP neural network is a type of feedforward artificial neural network that employs the backpropagation training method. It solves complex pattern recognition problems through the nonlinear transformation of hidden layers in the network. BP usually contains an input layer, one or more hidden layers, and an output layer, and it is able to learn the mapping relationship between input data and output data.

First, in this paper, *Sigmoid* function is used as the activation function to realize the nonlinear transformation of hidden layers as follows,

$$f(x) = \frac{1}{1 + e^{-x}} \quad (21)$$

Each node h_j of the hidden layer computes its output through the weighted summation of the input x_i and application of an activation function. The node O_m of the output layer is then a linear combination of the weighted summation of the nodes of the hidden layer and the bias b_m . The prediction error e_m is defined as the difference between the actual output y_m and the predicted output O_m ,

which is a key metric for network performance evaluation.

$$h_j = f\left(\sum_{i=1}^n W_{ij}x_i\right) \quad (22)$$

$$O_m = \sum_{j=1}^k h_j W_{jm} - b_m \quad (23)$$

$$e_m = y_m - O_m \quad (24)$$

To reduce the prediction error, the weights W_{ij} and bias b_m of the network are updated according to the error gradient. The weights are updated with

$$W_{ij} \leftarrow W_{ij} + \eta \frac{\partial e_m}{\partial W_{ij}} \quad (25)$$

The bias update formula is

$$b_m \leftarrow b_m + \eta \frac{\partial e_m}{\partial b_m} \quad (26)$$

where the learning rate η is a positive constant that controls the step size of the update. As a result, the BP neural network is able to continuously adjust its parameters during the iteration process to improve the accuracy of the prediction.

5 RNN

Recurrent Neural Network (RNN), as a neural network capable of processing sequential data, has a structure that consists of an input layer, a hidden layer, and an output layer. The unique feature of RNN is that the state of its hidden layer can be passed on to the next time step, thus maintaining the memory of the previous information. The dynamics of RNN can be described by the following set of equations.

The update of the hidden layer state is performed at time t through,

$$h_t = \phi(W^{in}x_t + W^{recur}h_{t-1} + b_h) \quad (27)$$

where the state of the hidden layer h_t is a function based on the current input x_t and the previous hidden state h_{t-1} , and b_h is the bias term of the hidden layer. W^{in} and W^{recur} are the input weight matrix and recursive weight matrix, respectively. The output y_{t+T} at time $t+T$ is computed from the current hidden state h_t and the bias b_y , with W^{out} being the output weight matrix

$$y_{t+T} = \psi(W^{out}h_t + b_y) \quad (28)$$

Here, ϕ and ψ represent the activation functions of the hidden and output layers, respectively, which can be *tanh*, *ReLU*, or *Sigmoid*, etc. In the option price prediction, the choice of activation functions should take into account the model's responsiveness to data and the characteristics of the prediction task. Option pricing models usually need to capture the nonlinear dynamics of the market and also avoid the problem of vanishing or exploding gradients, especially when dealing with time series data. In this paper, *tanh* is chosen as the activation function in RNN.

In this way, the RNN can effectively transfer the information in the input sequence to the output, which is suitable for a variety of application scenarios such as language modeling and speech recognition.

6 LSTM

LSTM, as an improvement of RNN, significantly enhances the processing of time-series data by means of a carefully designed gating mechanism. The core property of LSTM is its ability to capture and maintain long-term data dependencies due to three key structures within it: forgetting gates, input gates, and output gates. These gating units work together to determine the retention and updating of information, thus solving the gradient vanishing problem in traditional RNN. Based on Yu et al. [48], we can summarize the computational framework of LSTM in the following steps.

First, the computation of the forgetting gate F_t , which determines the information retained from the previous memory state, is performed through

$$F_t = \sigma(W_f \cdot X_t + U_f \cdot H_{t-1} + B_f) \quad (29)$$

where σ denotes the *Sigmoid* activation function. This is followed by the computation of the input gate I_t and the output gate O_t ,

$$I_t = \sigma(W_i \cdot X_t + U_i \cdot H_{t-1} + B_i) \quad (30)$$

$$O_t = \sigma(W_o \cdot X_t + U_o \cdot H_{t-1} + B_o) \quad (31)$$

which controls the new information incorporated at the current time step and determines the output of the final memory state. Both uses a *Sigmoid* layer to decide on the introduction of the new information, and a *tanh* layer to modulate the value of the new information.

Then, the calculation of the candidate memory state is performed, which represents the potential memory content of the current time step. The current memory state C_t is updated by combining the information from the forgetting gate and the input gate:

$$\tilde{C}_t = \tanh(W_c \cdot X_t + U_c \cdot H_{t-1} + B_c) \quad (32)$$

$$C_t = F_t * C_{t-1} + I_t * \tilde{C}_t \quad (33)$$

where $*$ denotes the element-by-element multiplication, W and U represent the weight matrices of the input and hidden states, respectively, B denotes the bias term, and X_t and H_{t-1} represent the current input vector and the hidden state vector of the previous time step, respectively.

Finally, the current hidden state H_t is generated to reflect the output gate control of the current memory state

$$H_t = O_t * \tanh(C_t) \quad (34)$$

With the above computational steps, LSTM can effectively process and memorize the key information in time series data, making it a powerful tool in the tasks of time series prediction and analysis.

7 GRU

GRU is an improvement over the traditional LSTM, which simplifies the network structure by combining the forgetting gate and the input gate into a single update gate. This innovation not only reduces the number of parameters of the model, but also improves the training efficiency of the model. According to Cahuantzi et al. [49], the core calculation of GRU can be summarized as the following steps.

First, the reset gate R_t needs to be calculated, which determines the amount of information transferred from the previous state h_{t-1} to the current state and is given by

$$R_t = \varphi(W_r[h_{t-1}, x_t] + b_r) \quad (35)$$

Then, the calculation of the update gate Z_t needs to be performed, which is responsible for regulating the introduction of new information

$$Z_t = \varphi(W_z[h_{t-1}, x_t] + b_z) \quad (36)$$

Afterwards, the candidate hidden state $\tilde{C}_t$ should be computed, representing the combination of the current input x_t and the previous state h_{t-1}

$$\tilde{C}_t = \tanh(W_c[R_t \odot h_{t-1} + 1, x_t] + b_c) \quad (37)$$

The update of the final hidden state h_t combines the previous hidden state with the current candidate hidden state:

$$h_t = (1 - Z_t) \odot h_{t-1} + Z_t \odot \tilde{C}_t \quad (38)$$

The output calculates y_t maps the final hidden state to the output space through the activation function φ

$$y_t = \varphi(W_o h_t) \quad (39)$$

Here, φ represents the dot product activation function, $\odot$ represents the Hadamard product, and W_r , W_z , W_c , W_o and b_r , b_z , b_c represent the weight matrix and bias term of different gates, respectively.

Through these dynamic calculation steps, the GRU model effectively captures the long-term dependence in the serial data, while maintaining the simplicity and computational efficiency of the model.

8 MLP

The MLP, a foundational architecture of feedforward neural networks, comprises an input stage, one or more intermediary hidden stages, and a final output stage. The network can effectively deal with complex data relations through the nonlinear transformation ability of its hidden layer(s). The excellent performance of MLP on nonlinear problems is attributed to its ability to approximate arbitrary continuous functions. According to Rajabi et al. [50], MLP can be summarized as follows:

$$F_j = \frac{1}{1 + e^{-(\sum_{i=1}^n W_{ij} \cdot x_i + b_j)}} \quad (40)$$

$$Y_k = g\left(\sum_{j=1}^n F_j \cdot W'_{jk} + b'_k\right) \quad (41)$$

where x_i represents the input feature, n is the number of input features, W_{ij} is the weight matrix between the input and the hidden layer, b_j is the bias term of the hidden layer, F_j is the activation function of the hidden layer, and W'_{jk} and b'_k denote the weights and biases connecting the hidden layer to the output layer, respectively. The function g represents the activation mechanism of the output layer. In this way, MLP is able to transform input data into output decisions, and its application in the field of data analysis and pattern recognition is promising.

Appendix 2: Optimal parameters of each model

Models	Parameters	Remarks
LASSO	alpha=0.01 max_iter=400 selection='random'	
SVR	C=0.01 epsilon=0.1 kernel='linear'	
Ridge	alpha=200 fit_intercept=True solver='svd'	
BP	hidden_size1=10 hidden_size2=10 num_epochs = 50 Network structure: class BPModel(nn.Module): def __init__(self, input_dim, hidden1, hidden2, output_dim): super(BPModel, self).__init__() self.fc1 = nn.Linear(input_dim, hidden1) self.fc2 = nn.Linear(hidden1, hidden2) self.fc3 = nn.Linear(hidden2, output_dim) self.relu = nn.ReLU() self.sigmoid = nn.Sigmoid() self.dropout = nn.Dropout(0.3) def forward(self, x): x = self.fc1(x) x = self.relu(x) x = self.dropout(x) x = self.fc2(x) x = self.relu(x) x = self.dropout(x) x = self.fc3(x) return x	Common parameters of neural network models learning_rate = 0.01 batch_size = 64 Adam optimizer output_dim = 1
RNN	hidden_size = 50 num_layers = 1 num_epochs = 40	
LSTM	hidden_size=50 num_layers=2 num_epochs = 60	
GRU	hidden1_size=70 num_layers=2 num_epochs = 40	
MLP	num_epochs = 60 Network structure: class MLPModel(nn.Module): def __init__(self, input_dim, output_dim): super(MLPModel, self).__init__() self.fc1 = nn.Linear(input_dim, 64) self.fc2 = nn.Linear(64, 32) self.fc3 = nn.Linear(32, 16) self.fc4 = nn.Linear(16, output_dim) self.relu = nn.ReLU() self.dropout = nn.Dropout(0.1) def forward(self, x): x = self.fc1(x) x = self.relu(x) x = self.dropout(x) x = self.fc2(x) x = self.relu(x) x = self.dropout(x) x = self.fc3(x) x = self.relu(x) x = self.dropout(x) x = self.fc4(x) return x	

Data availability

Data will be made available on request.

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